Classical Mechanics

Lecture 3: Least Action, Local Equations, and Symmetry

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Chapter 1

Least Action, Local Equations, and Symmetry

The laws of physics have many subjects but a remarkably common form. Newtonian particles, electromagnetic fields, gravitation, and even the fields of modern physics can all be described by an action principle. Thermodynamics looks different because it is statistical, but its microscopic degrees of freedom still obey dynamical laws of this kind.

The phrase laws of physics is being used specifically here. Mathematical identities such as $1 + 1 = 2$, and the internal rules of probability theory, are not dynamical laws of a physical system. The fact that probability describes nature may itself be a physical fact, but the laws at issue now are those that govern motion and fields.

This is a stronger claim than saying that least action is a clever alternative to $F = ma$. The action is the organizing language; Newton's equations are one local expression of it. Our first task is therefore to understand how a global statement about an entire history can become a differential equation at each instant of time. Two elementary facts from calculus will do nearly all of the work.

1.1 Two Small Theorems

1.1.1 Integration by parts

Begin with the fundamental theorem of calculus,

$$\int_{t_1}^{t_2} \dot{F}(t)\, dt = F(t_2) - F(t_1). \tag{1.1}$$

There is a useful elementary picture behind this formula. Divide the interval into short steps. Each contribution to the integral is approximately a difference,

$$F(t_{k+1}) - F(t_k).$$

On adding the differences, every intermediate value cancels:

$$(F_2 - F_1) + (F_3 - F_2) + \cdots + (F_N - F_{N-1}) = F_N - F_1.$$

Only the two endpoints survive.

Now take $F = fg$. The product rule gives

$$\frac{d}{dt}(fg) = \dot{f}\,g + f\,\dot{g}. \tag{1.2}$$

Substitution into Eq. (1.1) yields

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = [fg]_{t_1}^{t_2} - \int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{1.3}$$

Thus a derivative may be moved from one factor to the other, at the cost of a minus sign and a boundary term.

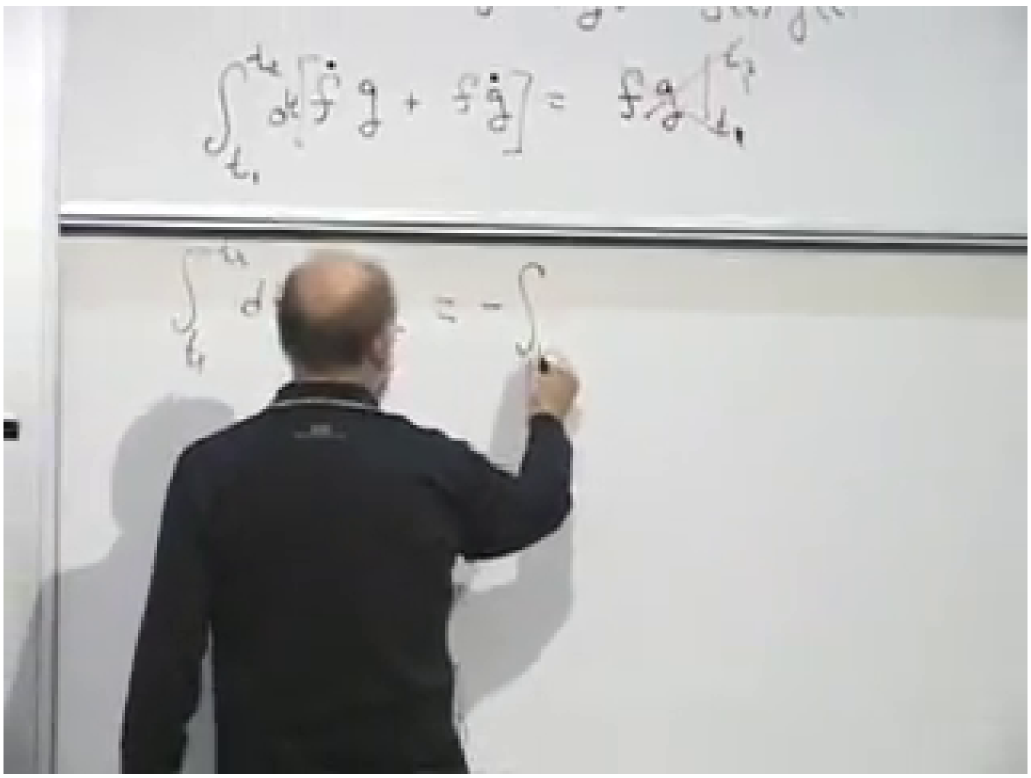

Figure 1.1: Integration by parts on the board at 00:10:20. The upper line retains the endpoint term; the lower line is being reduced to the endpoint-vanishing case used in the variation.

Lecture frame: 00:10:20.

Question from the audience. Why must the integral contain dt?

Response. An integral is always taken with respect to a variable. Here time is the independent variable, so the complete expression is $\int \dot{f}(t)\,dt$. Writing only $\int \dot{f}(t)$ leaves the integration measure unstated. Small notational questions are worth stopping for because they force every ingredient of the calculation into view.

We shall use a special case repeatedly. If f vanishes at t_1 and t_2, then $[fg]_{t_1}^{t_2} = 0$, and

$$\int_{t_1}^{t_2} \dot{f}\,g\,dt = -\int_{t_1}^{t_2} f\,\dot{g}\,dt. \tag{1.4}$$

It is enough that the product fg have the same value at both endpoints, but fixed-endpoint variations will make f vanish separately at each end.

1.1.2 An arbitrary test function

The second fact is almost obvious, but it is the step that turns an integral condition into a local law.

Proposition 1.1. *If*

$$\int_{t_1}^{t_2} a(t)f(t)\,dt = 0 \tag{1.5}$$

for every sufficiently smooth function $f(t)$ that vanishes at the endpoints, then $a(t) = 0$ throughout the interval.

Proof. Choose f to be a narrow smooth blip around an arbitrary point t_* and zero away from that neighborhood. If $a(t_*) \neq 0$, choose the sign of the blip to agree with the sign of a; for a sufficiently narrow neighborhood the integral cannot vanish. Hence $a(t_*) = 0$. Since the point was arbitrary, a vanishes everywhere. □

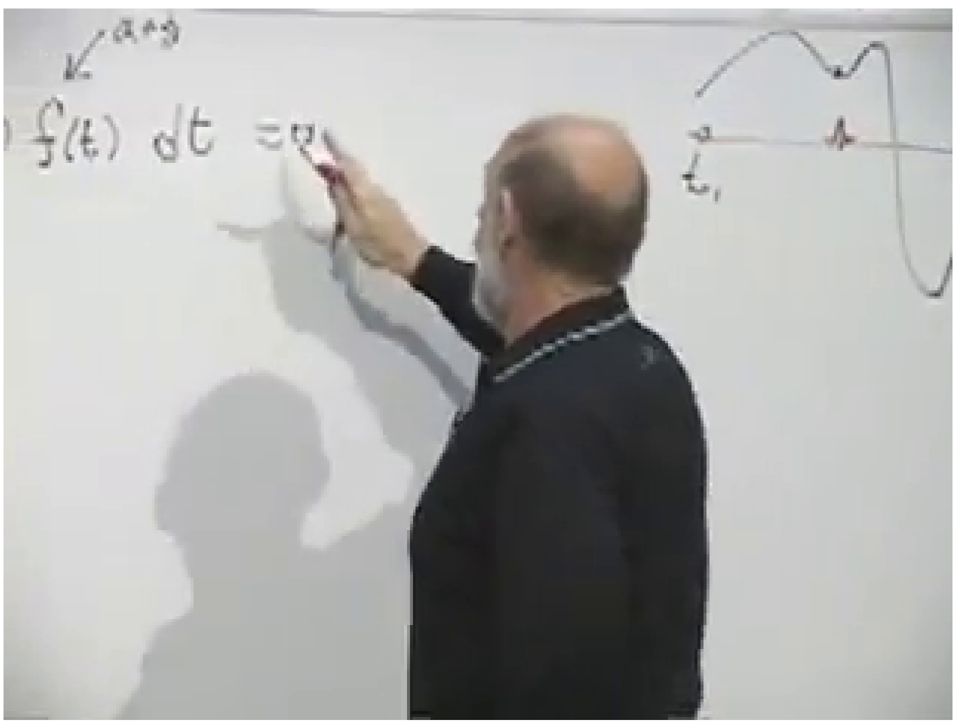

Figure 1.2: The arbitrary-function argument at 00:13:30. A narrow test function isolates the value of $a(t)$ near any chosen point of the interval.

Lecture frame: 00:13:30.

One does not verify Eq. (1.5) by trying functions one after another; that would take forever. In an application we prove the integral vanishes for an arbitrary f, and the proposition then gives the pointwise conclusion in one step.

1.2 Histories: Local and Global Descriptions

Let a system have generalized coordinates

$$q_1(t), q_2(t), \ldots, q_n(t). \tag{1.6}$$

The complete collection of these functions is a *history*, or a trajectory in configuration space. For a system of N unconstrained particles in three dimensions, $n = 3N$.

Whether time is drawn horizontally, as the independent variable often is in elementary mathematics, or vertically, as it often is in physics diagrams, is only a convention. The object represented is the same set of functions $q_i(t)$.

There are two apparently different ways to state a law of motion. A local law uses information near one instant to continue the trajectory. In Newtonian mechanics, knowledge of q_i and $\dot{q}_i$, together with equations of the form $F = ma$, determines the acceleration and advances the solution to the next instant.

A global law considers the whole path between two endpoint configurations. It assigns a number, the action, to every candidate history and says that the physical history makes that number stationary. Instead of giving q_i and $\dot{q}_i$ at one time, this boundary-value formulation fixes q_i at an initial and a final time.

The two formulations must be equivalent when they describe the same physics. There is already a simple reason to expect locality to emerge. If a path is stationary on a long interval, its segment between any two neighboring points must also be stationary with those points held fixed. Otherwise that segment could be replaced and the action of the whole path would change to first order.

1.3 Varying a Path

Write the physical history as $\hat{q}_i(t)$. A nearby history can be parameterized by

$$q_i(t, \alpha) = \hat{q}_i(t) + \alpha f_i(t), \qquad f_i(t_1) = f_i(t_2) = 0. \tag{1.7}$$

The number α controls the size and sign of the deformation; it does not depend on time. The functions $f_i(t)$ describe its shape and are otherwise arbitrary. The endpoint conditions nail the beginning and end of every candidate path in place.

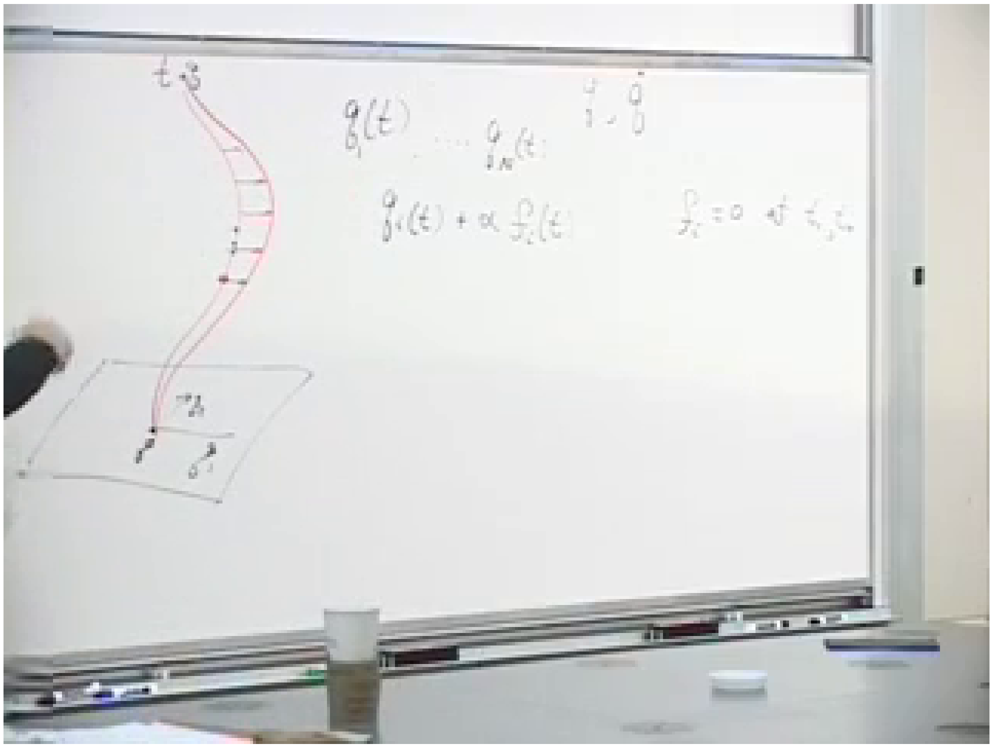

Figure 1.3: A fixed-endpoint deformation at 00:22:55. The red curve is displaced in the interior while the endpoint conditions $f_i(t_1) = f_i(t_2) = 0$ remain visible on the board.

Lecture frame: 00:22:55.

For fixed $\hat{q}_i$ and fixed deformation functions f_i, the action is now an ordinary function $A(\alpha)$. The physical path corresponds to $\alpha = 0$. If it is a minimum, then in particular it is stationary:

$$\left.\frac{dA}{d\alpha}\right|_{\alpha=0} = 0. \tag{1.8}$$

The first derivative is the condition we need; whether the stationary path is a minimum, maximum, or saddle requires additional information.

Question from the audience. What is the action being varied?

Response. For a history between t_1 and t_2, the action is

$$A[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt. \tag{1.9}$$

The Lagrangian L is evaluated at every point of the path. In the Newtonian examples below it is $L = T - U$, but the variational calculation needs only that L be a function of the coordinates, velocities, and possibly time.

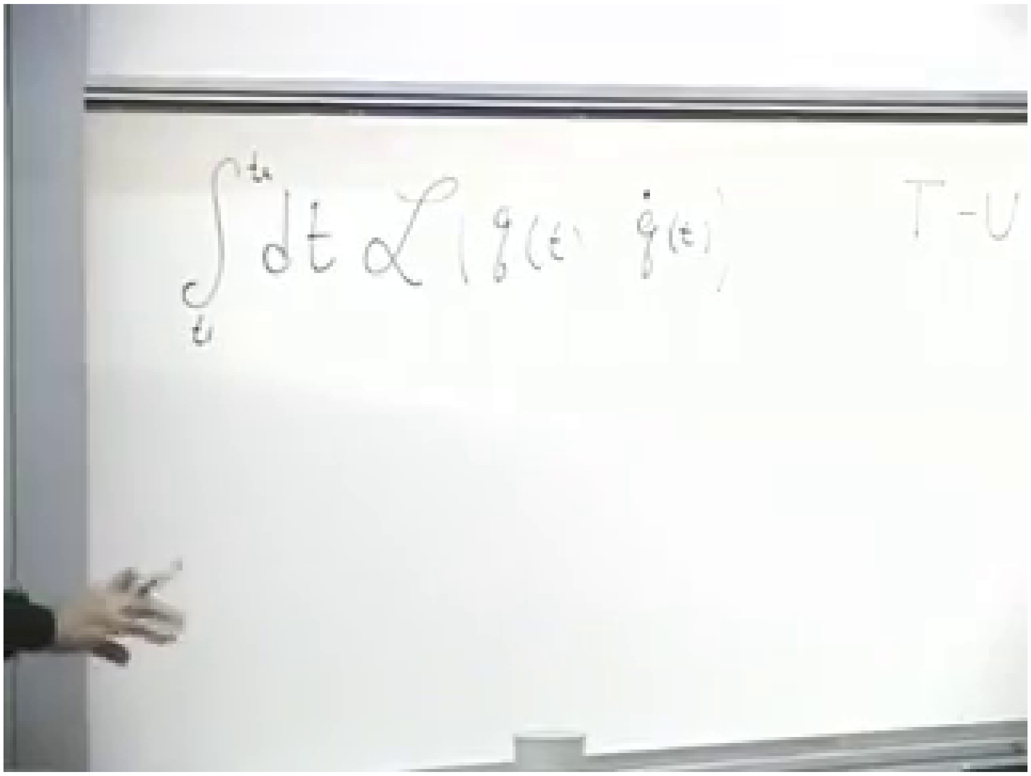

Figure 1.4: The action integral at 00:28:42, written before any particular mechanical form of the Lagrangian is needed.

Lecture frame: 00:28:42.

Equation (1.7) immediately gives

$$\frac{\partial q_i}{\partial \alpha} = f_i, \qquad \frac{\partial \dot{q}_i}{\partial \alpha} = \dot{f}_i. \tag{1.10}$$

Applying the chain rule inside the action,

$$\frac{dA}{d\alpha} = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} \frac{\partial q_i}{\partial \alpha} + \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{q}_i}{\partial \alpha} \right) dt \tag{1.11}$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} f_i + \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \right) dt. \tag{1.12}$$

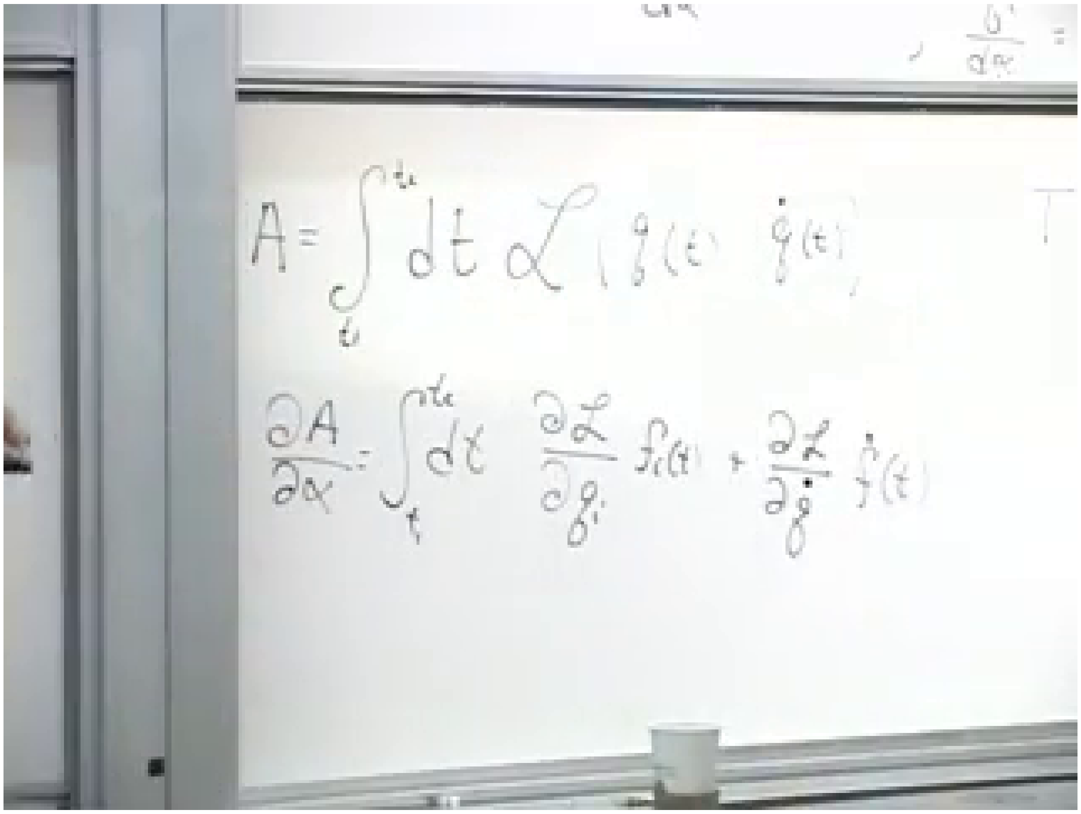

Figure 1.5: The first variation at 00:34:10. Changes of both q_i and $\dot{q}_i$ contribute, and the audience supplies the required sum over coordinates.

Lecture frame: 00:34:10.

Question from the audience. Should there be a sum over i?

Response. Yes. Each coordinate and its velocity are independent arguments of L, so the chain rule contains one pair of terms for every i. For a one-coordinate problem the sum is invisible; for a general system it is essential.

The unwanted $\dot{f}_i$ in the second term is precisely why integration by parts was prepared. Applying Eq. (1.3),

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_i} \dot{f}_i \, dt = \left[f_i \frac{\partial L}{\partial \dot{q}_i} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} f_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) dt. \tag{1.13}$$

In the earlier notation, f_i plays the role of f, while $\partial L / \partial \dot{q}_i$ plays the role of g. The operation removes the dot from f_i and differentiates the momentum-like factor instead. The boundary term vanishes because $f_i(t_1) = f_i(t_2) = 0$. Consequently,

$$\left. \frac{dA}{d\alpha} \right|_{\alpha=0} = \int_{t_1}^{t_2} \sum_i \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \right]_{\hat{q}} f_i(t) \, dt. \tag{1.14}$$

Question from the audience. What happened to the endpoint contribution?

Response. It is exactly $[f_i \, \partial L / \partial \dot{q}_i]_{t_1}^{t_2}$. The varied and physical paths have the same endpoints, so every f_i vanishes there and the contribution is zero. The endpoint condition was chosen at the start for this reason.

Stationarity requires Eq. (1.14) to vanish for every collection of functions $f_i(t)$. The arbitrary-function proposition applies separately to every i, giving

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}} \qquad (i = 1, \ldots, n). \tag{1.15}$$

These are the Euler–Lagrange equations. The global condition $\delta A = 0$ has become one local differential equation for each generalized coordinate.

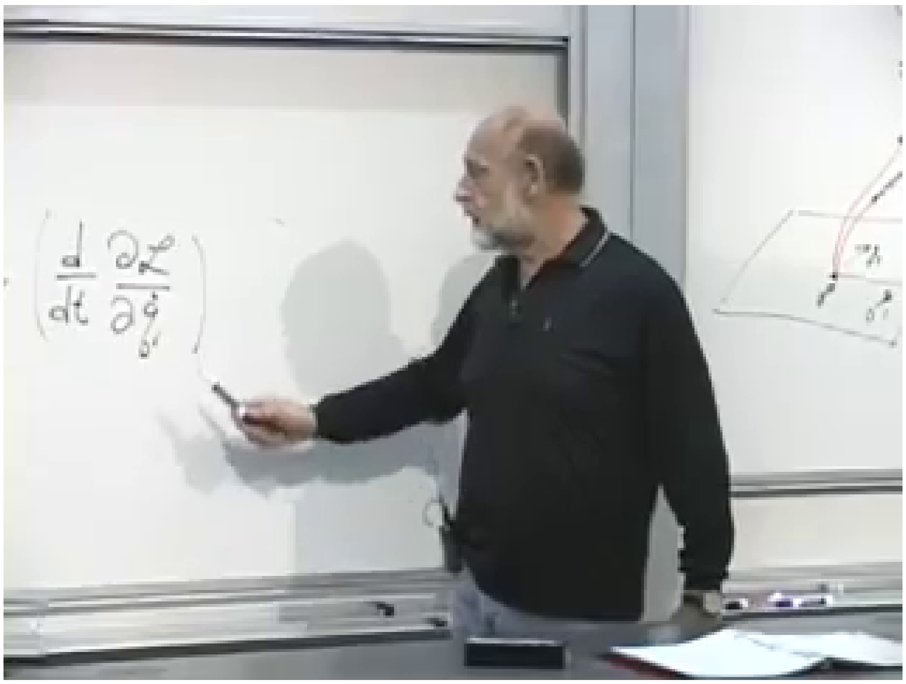

Figure 1.6: The Euler–Lagrange time-derivative term at 00:45:02, beside the trajectory picture used to connect the global path with its local equation.

Lecture frame: 00:45:02.

The result is not confined to point particles. Field theory, general relativity, and string theory use the same variational structure with more elaborate coordinates. Quantum mechanics also retains it in another form: Feynman's path integral assigns amplitudes to histories through their action.

1.4 Momentum and Generalized Force

The quantity differentiated with respect to time in Eq. (1.15) has a name:

$$\pi_i \equiv \frac{\partial L}{\partial \dot{q}_i}. \tag{1.16}$$

It is the canonical momentum conjugate to q_i. The corresponding $\partial L/\partial q_i$ is the generalized force. The equation of motion therefore reads

$$\dot{\pi}_i = \frac{\partial L}{\partial q_i}. \tag{1.17}$$

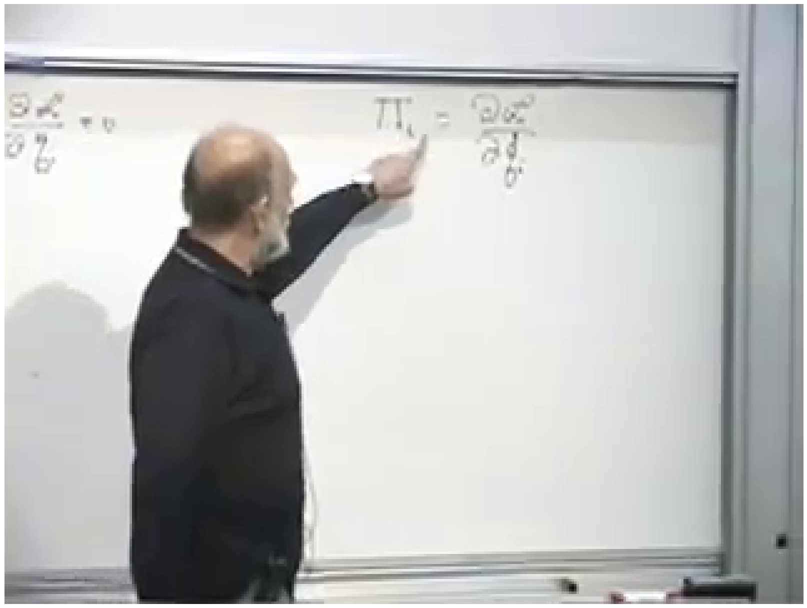

Figure 1.7: Canonical momentum introduced on the board at 00:46:32. The standard lowercase notation is $\pi_i = \partial L/\partial \dot{q}_i$.

Lecture frame: 00:46:32.

Question from the audience. What does the Euler–Lagrange equation say in ordinary language?

Response. First, it says that a global statement about the entire trajectory is equivalent to a local differential law. Second, after naming the two sides, it says that the time derivative of momentum equals generalized force. The concrete examples now show that this is $F = ma$, written in a form that survives arbitrary choices of coordinates.

One may try to picture the left side as a time change of the velocity-derivative of L, and the right side as the coordinate-derivative at the present point. That intuition is useful, but the names momentum and force become convincing only after the first calculation.

1.5 Recovering Newton's Equation

For one particle of mass m moving on a line,

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - U(x). \tag{1.18}$$

The generalized coordinate q is simply x, and its canonical momentum is the ordinary momentum:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} = p. \tag{1.19}$$

The coordinate derivative is

$$\frac{\partial L}{\partial x} = -\frac{dU}{dx}. \tag{1.20}$$

Hence

$$m\ddot{x} = -\frac{dU}{dx}, \tag{1.21}$$

which is Newton's law for a conservative force.

The minus sign is not optional. Had we chosen $L = T + U$, the force would have come out with the wrong sign. The Lagrangian $T - U$ is not the energy and is not generally conserved along a trajectory. For a time-independent conservative system the familiar conserved energy is $T + U$; its derivation from symmetry is deliberately postponed.

For many particles, combine every particle label and Cartesian direction into one index i. Then

$$L = \sum_i \frac{1}{2} m_i \dot{x}_i^2 - U(x_1, \ldots, x_n), \tag{1.22}$$

and, coordinate by coordinate,

$$p_i = m_i \dot{x}_i, \qquad \dot{p}_i = -\frac{\partial U}{\partial x_i}. \tag{1.23}$$

For a physical particle the same mass occurs in its three Cartesian components, although the notation is general enough to attach a coefficient to every coordinate. A rigid body, an eraser, or a person can ultimately be viewed as a large collection of particles held together by internal forces, so this construction already covers an enormous class of mechanical systems.

1.6 Translation Symmetry and Momentum

Consider two interacting particles on a line:

$$L = \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 - U(x_1 - x_2). \tag{1.24}$$

The potential depends only on their separation. If both coordinates are shifted by the same constant a,

$$x_1 \longmapsto x_1 + a, \qquad x_2 \longmapsto x_2 + a, \tag{1.25}$$

their velocities and the difference $x_1 - x_2$ are unchanged. The Lagrangian, and therefore the action, is invariant. This is translation symmetry.

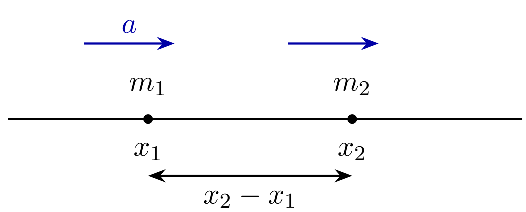

Figure 1.8: A rigid translation moves both particles by the same amount without changing their separation or velocities.

This form would fail if a third, external lump of matter exerted a position-dependent force. The potential would then depend on the distance from that external object as well as on the separation between the two particles. The two-particle subsystem would no longer be closed or translation invariant.

Set $d = x_1 - x_2$. The chain rule gives

$$\frac{\partial U}{\partial x_1} = \frac{dU}{dd}, \qquad \frac{\partial U}{\partial x_2} = -\frac{dU}{dd}. \tag{1.26}$$

Therefore

$$\dot{p}_1 = -\frac{\partial U}{\partial x_1}, \qquad \dot{p}_2 = -\frac{\partial U}{\partial x_2} = +\frac{\partial U}{\partial x_1}. \tag{1.27}$$

The forces are equal and opposite, and adding the equations gives

$$\frac{d}{dt}(p_1 + p_2) = 0. \tag{1.28}$$

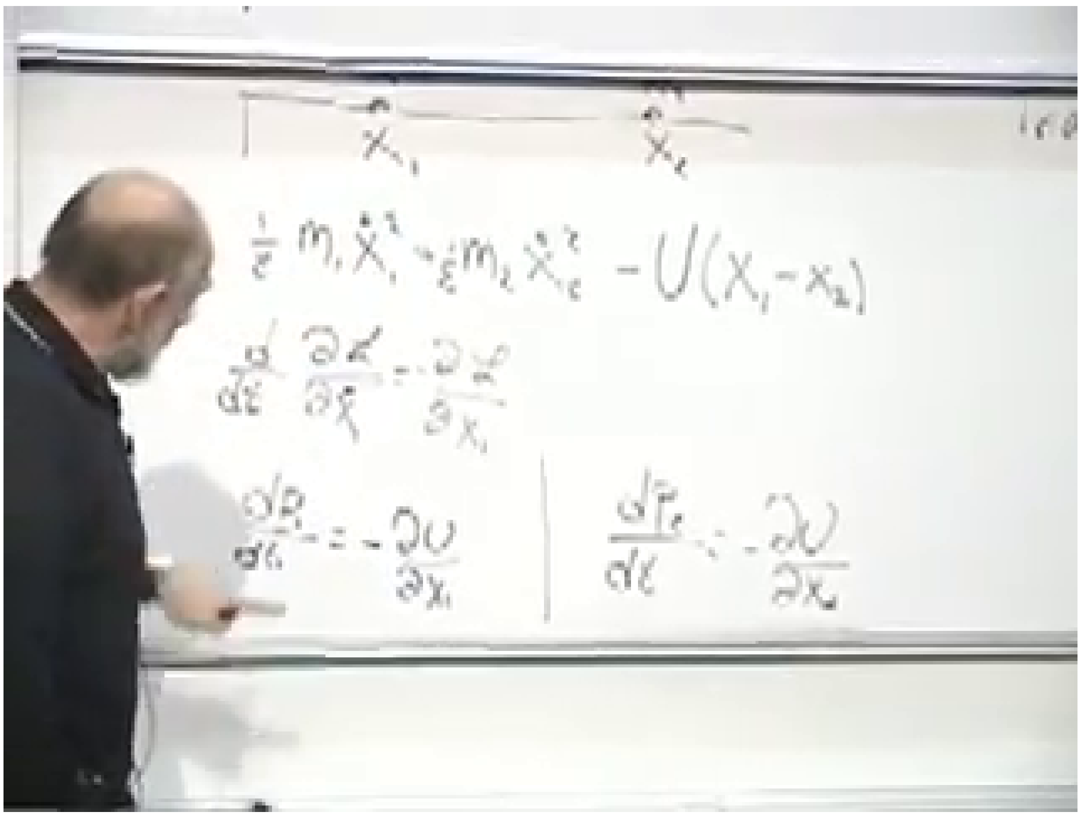

Figure 1.9: The translation-invariant Lagrangian and its two force equations at 01:08:50. Dependence on $x_1 - x_2$ makes the derivatives equal and opposite.

Lecture frame: 01:08:50.

This is the first instance of the central pattern:

$$\text{translation symmetry} \quad \Longrightarrow \quad \text{conservation of linear momentum.}$$

The symmetry is an active operation on the system that leaves the action unchanged. Here the operation can be made by an arbitrarily small amount and built up continuously, so it is a continuous symmetry.

Question from the audience. If U depends on $|x_1 - x_2|$, is inversion another symmetry?

Response. Yes. Interchanging the orientation of the separation leaves such a potential unchanged. Unlike translation, inversion is discrete: there is no continuously adjustable parameter connecting the two alternatives. Continuous symmetries generate the classical conservation laws being studied here. Discrete symmetries become especially consequential in quantum mechanics, which we leave for later.

1.7 One Translation Survives

Place a particle near Earth's surface and retain horizontal x and vertical y motion. With y measured upward,

$$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy, \tag{1.29}$$

where g is the approximately constant local gravitational acceleration. Numerically it is about $9.8\,\mathrm{m\,s^{-2}}$, commonly rounded to $10\,\mathrm{m\,s^{-2}}$ for this discussion.

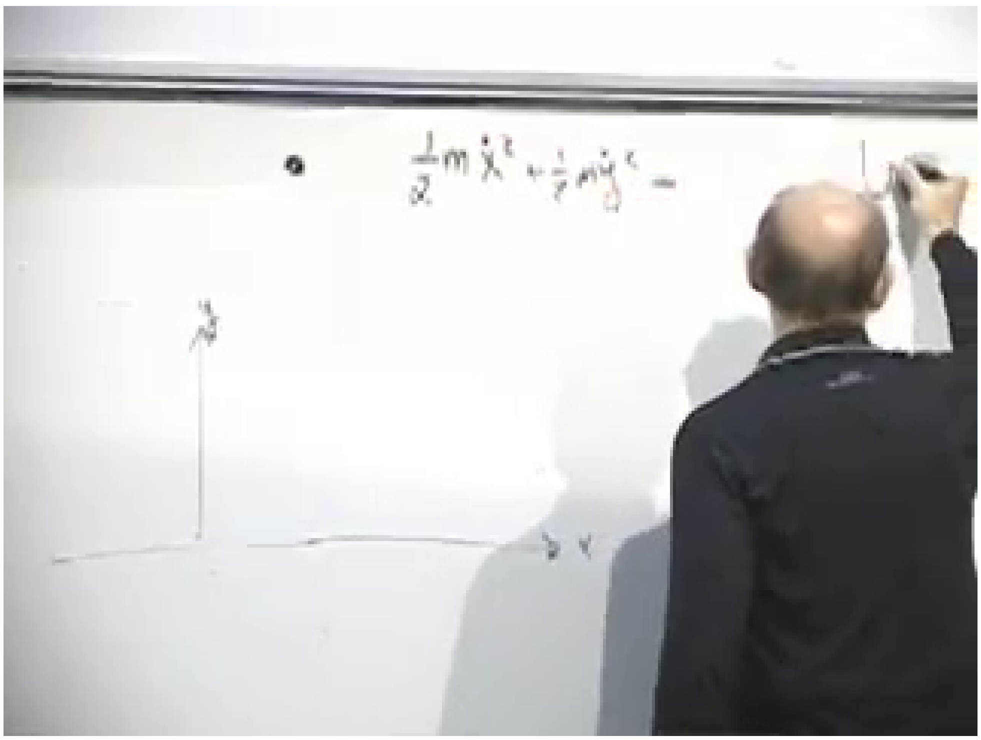

Figure 1.10: The two-coordinate kinetic energy and near-Earth setup at 01:15:15. The stable board content is $T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2$; the potential is completed immediately afterward.

Lecture frame: 01:15:15.

The Lagrangian contains no x, so horizontal translation is a symmetry:

$$\frac{d}{dt}(m\dot{x}) = 0. \tag{1.30}$$

It does contain y, so vertical translation is not a symmetry:

$$\frac{d}{dt}(m\dot{y}) = -mg, \qquad \ddot{y} = -g. \tag{1.31}$$

There is one translation symmetry and one corresponding conserved momentum, not two.

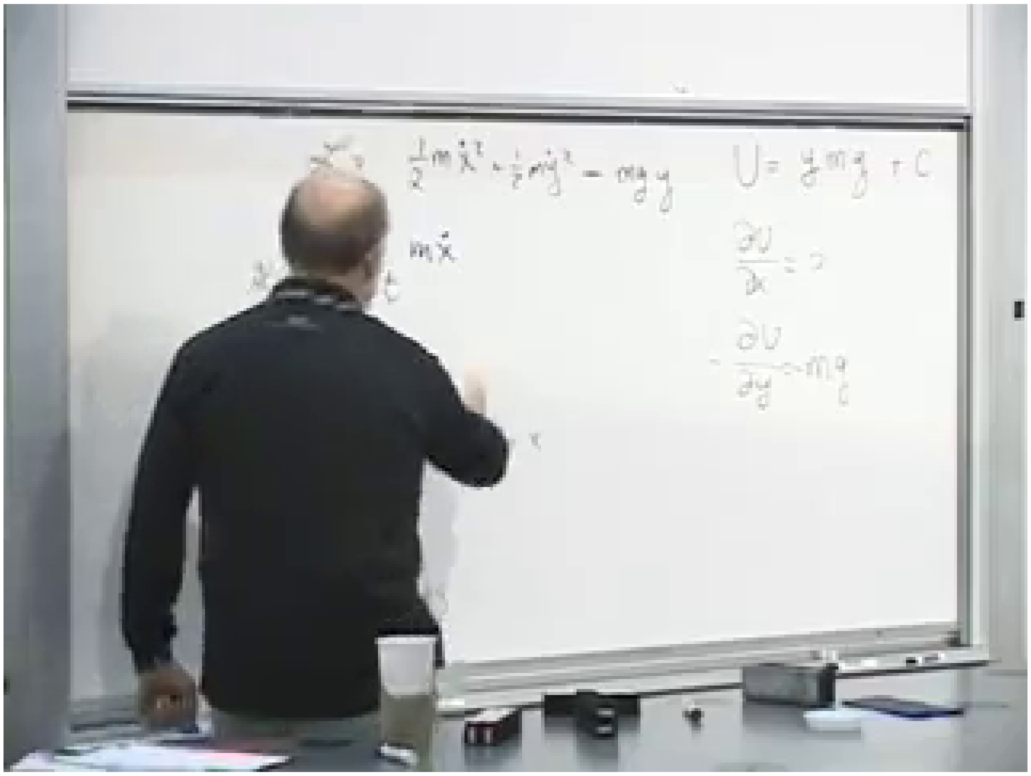

Figure 1.11: The completed near-Earth Lagrangian and coordinate derivatives at 01:19:15. The absence of x gives horizontal momentum conservation, while $\partial U/\partial y = mg$ gives downward acceleration.

Lecture frame: 01:19:15.

Question from the audience. For a very small vertical displacement, is there an approximate y-translation symmetry?

Response. No exact symmetry appears merely because the displacement is small. In the near-Earth approximation $U = mgy$, any nonzero change of y changes the potential by $mg\,\Delta y$. We may add one fixed constant to U and thereby choose a different zero of potential energy, but that does not remove the y-dependence or the vertical force.

Adding C to the potential changes neither force nor trajectory:

$$U(y) \longmapsto U(y) + C, \qquad -\frac{d}{dy}(U + C) = -\frac{dU}{dy}. \tag{1.32}$$

Sea level, the floor, or any other fixed height may therefore be chosen as the zero of potential energy.

Question from the audience. Is this just the familiar falling-body problem and its energy conservation in another form?

Response. It is the same mechanics, but so far we have derived the equations of motion and identified momentum conservation. The conserved energy is $T + U$, not the Lagrangian $T - U$. Its origin will be traced to another symmetry, time translation, in a later step.

1.8 Changing Coordinates

The action does not depend on how we label the configurations. Rectangular, polar, parabolic, or curved coordinates may make the formulas look different, but a correctly transformed action assigns the same number to the same physical history. Stationarity of that action is therefore coordinate-independent.

For planar motion, replace x, y by polar coordinates r, θ. At every instant the velocity has a radial component and a perpendicular component:

$$v_r = \dot{r}, \qquad v_\perp = r\dot{\theta}. \tag{1.33}$$

The factor of r has a simple geometric origin. A fixed angular change $d\theta$ sweeps the arc length $r\,d\theta$, so the same angular speed corresponds to a larger linear speed farther from the origin. The fingertips of a rotating arm move faster than the elbow.

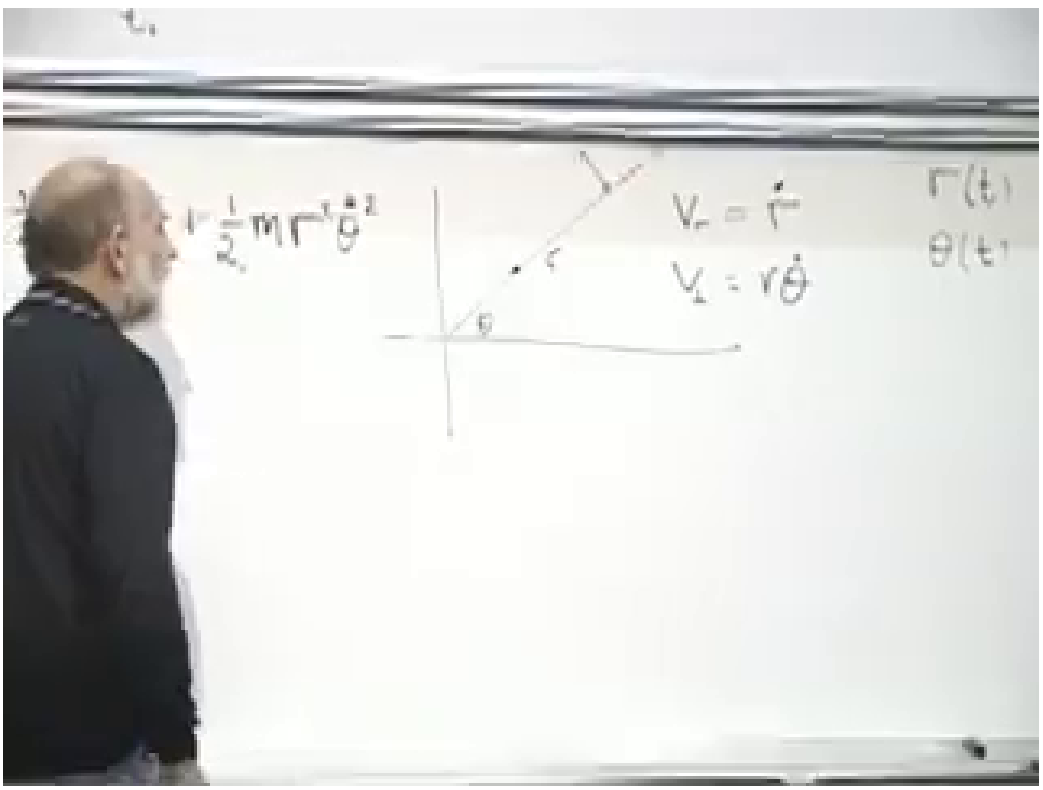

Figure 1.12: Polar-coordinate velocity components at 01:27:50. The board records $v_r = \dot{r}$, $v_\perp = r\dot{\theta}$, and the resulting kinetic-energy terms.

Lecture frame: 01:27:50.

Hence

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right). \tag{1.34}$$

For a central potential $U(r)$,

$$L = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 - U(r). \tag{1.35}$$

It is useful to write both Euler–Lagrange equations. The radial one is

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dU}{dr}. \tag{1.36}$$

The $mr\dot{\theta}^2$ term comes from differentiating the angular kinetic energy with respect to r. Radial momentum is not conserved: the central force and the changing radial direction both enter its evolution.

The angular coordinate is different. The Lagrangian depends on $\dot{\theta}$ but not on θ itself. Thus θ is a cyclic coordinate and

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}. \tag{1.37}$$

Its equation of motion is

$$\frac{d}{dt}\left(mr^2\dot{\theta}\right) = \frac{\partial L}{\partial \theta} = 0. \tag{1.38}$$

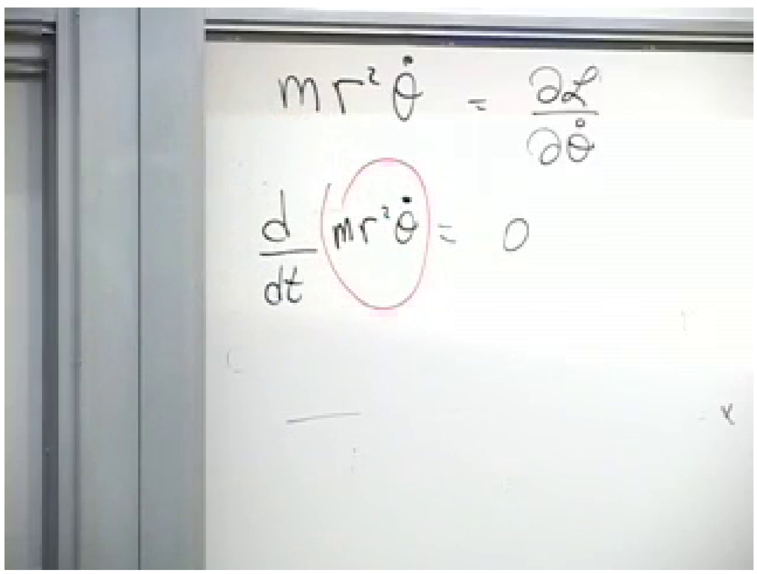

Figure 1.13: Angular momentum conservation completed on the board at 01:33:35: $\frac{d}{dt}(mr^2\dot{\theta}) = 0$.
Lecture frame: 01:33:35.

The conserved quantity $mr^2\dot{\theta}$ is the angular momentum, usually denoted by ℓ when L is already reserved for the Lagrangian. Solving for the angular speed,

$$\dot{\theta} = \frac{\ell}{mr^2}. \tag{1.39}$$

As r decreases, $\dot{\theta}$ increases. This is the spinning-skater effect in its simplest form.

We have now seen the same structure twice:

$$\begin{aligned} \text{translation invariance} &\implies \text{linear momentum conservation,} \\ \text{rotational invariance} &\implies \text{angular momentum conservation.} \end{aligned}$$

The action principle makes the pattern visible. A coordinate that does not appear in the Lagrangian has a conserved conjugate momentum, and the local equation expressing that fact is itself the consequence of stationarity of the entire history.